Real-Gas Property Software Based on the Peng–Robinson Equation of State: PVT Behaviour, Residual Enthalpy, Entropy and Fugacity

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Core Objective

The aim of this project is to **predict how a real gas behaves at a given temperature and pressure** using the Peng–Robinson (PR) cubic equation of state, implemented as a software tool written in Dart with the Flutter framework. The ideal-gas law $PV = RT$ fails at high pressure and low temperature, where intermolecular attraction and the finite size of molecules matter. For a chosen substance, the software computes:

(a) **PVT behaviour**: the compressibility factor $Z = PV/RT$, molar volume and all real roots of the cubic, with any two of T, P and V as inputs.

(b) **Residual properties** H^R and S^R: how far the real-fluid enthalpy and entropy are from the ideal-gas values at the same T and P, plus the absolute H and S relative to an ideal-gas reference state.

(c) **Fugacity** f and **fugacity coefficient** ϕ, which are used to pick the stable phase (vapour, liquid or supercritical) when the cubic has three roots.

The results are also shown as charts (a generalised compressibility chart, P–V isotherms, H–T and S–T curves, and ϕ, H^R, S^R against pressure) and can be exported as a PDF report.

Theory: the Peng–Robinson Equation of State

The PR equation is

$$P = \frac{RT}{V-b} - \frac{a(T)}{V^2 + 2bV - b^2}$$

where the parameters come from the critical constants T_c, P_c and the acentric factor ω:

$$a(T) = 0.45724\,\frac{R^2 T_c^2}{P_c}\,\alpha(T_r), \qquad b = 0.07780\,\frac{RT_c}{P_c}, \qquad T_r = \frac{T}{T_c}$$

$$\alpha(T_r) = \left[1 + \kappa\left(1 - \sqrt{T_r}\right)\right]^2, \qquad \kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2$$

The factor α makes the attraction term depend on temperature and molecular shape (through ω), which is why PR predicts vapour pressures and liquid densities better than van der Waals or Redlich–Kwong. **Cubic in Z.** Substituting $V = ZRT/P$ and defining the dimensionless groups

$$A = \frac{aP}{(RT)^2}, \qquad B = \frac{bP}{RT}$$

gives the polynomial that the software solves:

$$Z^3 - (1-B)\,Z^2 + \left(A - 3B^2 - 2B\right)Z - \left(AB - B^2 - B^3\right) = 0$$

Only roots with $Z > B$ (that is, $V > b$) are physical. If there is one real root, the fluid is single-phase. Below T_c there can be three real roots: the largest is the vapour root, the smallest is the liquid root, and the middle one lies on the unstable part of the isotherm and is discarded.

Derivation of H^R, S^R and $\ln\phi$ for Peng–Robinson

A residual property is the real-fluid value minus the ideal-gas value at the same T and P: $M^R = M - M^{ig}$. For a pressure-explicit equation the cleanest route is the residual Helmholtz energy at fixed T and V, since $P(T,V)$ integrates directly.

Step 1: residual Helmholtz energy. From $(\partial A/\partial V)_T = -P$, integrating from infinite volume (where every fluid is ideal) to V:

$$A^R(T,V) = -\int_\infty^V \left(P - \frac{RT}{V}\right) dV$$

In the PR integrand, the repulsive part $\frac{RT}{V-b} - \frac{RT}{V}$ integrates to $RT\ln\frac{V-b}{V}$. For the attractive part, $V^2 + 2bV - b^2 = \Big(V + (1+\sqrt{2})b\Big)\Big(V + (1-\sqrt{2})b\Big)$, whose roots differ by $2\sqrt{2}\,b$, so by partial fractions

$$\int_\infty^V \frac{dV}{\Big(V + (1+\sqrt{2})b\Big)\Big(V + (1-\sqrt{2})b\Big)} = \frac{1}{2\sqrt{2}\,b}\ln\frac{V + (1-\sqrt{2})b}{V + (1+\sqrt{2})b} = -\frac{L}{2\sqrt{2}\,b}$$

where the logarithmic term used throughout the software is

$$L = \ln\frac{V + (1+\sqrt{2})\,b}{V + (1-\sqrt{2})\,b} = \ln\frac{Z + (1+\sqrt{2})\,B}{Z + (1-\sqrt{2})\,B}$$

Combining, and using $\frac{V-b}{V} = \frac{Z-B}{Z}$:

$$A^R(T,V) = -RT\ln\frac{Z-B}{Z} - \frac{a}{2\sqrt{2}\,b}L$$

Step 2: convert from fixed (T, V) to fixed (T, P). The ideal gas at the same T and P occupies $V^{ig} = RT/P = V/Z$, not V. Correcting for this volume difference adds $RT\ln Z$ to the Helmholtz and Gibbs energies and $-R\ln Z$ to the entropy, and $G = A + PV$ adds $RT(Z-1)$:

$$G^R(T,P) = A^R(T,V) + RT(Z-1) - RT\ln Z$$

Step 3: fugacity coefficient. Since $\ln\phi = G^R/RT$, and $\frac{a}{bRT} = \frac{A}{B}$:

$$\ln\phi = \frac{G^R}{RT} = Z - 1 - \ln(Z-B) - \frac{A}{2\sqrt{2}\,B}\ln\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B}, \qquad f = \phi P$$

Step 4: residual entropy. $S = -(\partial A/\partial T)_V$. At fixed V, Z changes with T but L depends only on V/b, so only $a(T)$ is differentiated in the attractive term. Adding the $R\ln Z$ correction from Step 2:

$$S^R(T,P) = -\left(\frac{\partial A^R}{\partial T}\right)_V + R\ln Z = R\ln(Z-B) + \frac{1}{2\sqrt{2}\,b}\frac{da}{dT}L$$

Step 5: residual enthalpy. From $H^R = G^R + TS^R$, the $\ln(Z-B)$ terms cancel:

$$H^R(T,P) = RT(Z-1) + \frac{T\dfrac{da}{dT} - a}{2\sqrt{2}\,b}L$$

Step 6: temperature derivative of a. With $a = a_c\,\alpha$ and $\sqrt{\alpha} = 1 + \kappa(1 - \sqrt{T_r})$:

$$\frac{d\alpha}{dT} = 2\sqrt{\alpha}\,\frac{d\sqrt{\alpha}}{dT} = -\kappa\sqrt{\frac{\alpha}{T\,T_c}}, \qquad \frac{da}{dT} = -a_c\,\kappa\sqrt{\frac{\alpha}{T\,T_c}}$$

Since $da/dT < 0$ and $L > 0$, H^R and S^R are negative for a real gas, and all three properties go to zero as $P \to 0$ ($B \to 0$, $Z \to 1$, $L \to 0$), a limit the software's charts satisfy.

Absolute enthalpy and entropy. The software reports H and S relative to the ideal gas at $T_{ref} = 298.15$ K and $P_{ref} = 1$ bar (where $H = 0$, $S = 0$), by adding the ideal-gas heat-capacity integral to the residual term:

$$H(T,P) = \int_{298.15}^{T} C_P^{ig}\,dT + H^R(T,P), \qquad S(T,P) = \int_{298.15}^{T} C_P^{ig}\,\frac{dT}{T} - R\ln\frac{P}{1\text{ bar}} + S^R(T,P)$$

with $C_P^{ig}/R = A + BT + CT^2 + DT^{-2}$ (SVNA Table C.1). Only differences between two states are physically meaningful; a jump in the H–T or S–T curve marks a phase change.

Program Structure and Algorithms

The software is written in **Dart with the Flutter framework**, with the physics kept separate from the user interface:

File (Dart module)	What it does
`substances.dart`	Database of T_c, P_c, ω, molar mass (SVNA Table B.1): methane, ethane, ethylene, propane, n-butane, CO_2, N_2, O_2, NH_3, H_2S.
`pr_eos.dart`	Core solver: κ, α, a, b, da/dT, A, B; solves the cubic, keeps roots with $Z > B$; evaluates H^R, S^R, $\ln\phi$, f, H, S; labels the phase.
`main.dart`	Interface: substance picker, input mode, validated inputs, *Calculate*, results panel, Guide page, light/dark theme.
`chart_data.dart`	Sweeps P, T or V through the solver to build plot data, with ideal-gas curves for comparison.
`chart_painter.dart`	Custom painter that draws the axes, curves and labels.
`pdf_report.dart`	Exports a "Peng–Robinson EOS Report" PDF of inputs, equations and results to save, print or share.

Input modes. The user chooses which two state variables are known:

1. *Given T and P*: solve the cubic for Z, then $V = ZRT/P$.
2. *Given T and V*: PR is explicit in pressure, so P follows directly from the equation; the software rejects $V \le b$ and any state where the computed P is not positive (inside the unstable spinodal region).
3. *Given P and V*: T appears inside $\alpha(T)$, so the equation is solved numerically for T.

Phase and root selection. When three physical roots exist, the software computes $\ln\phi$ for the vapour and liquid roots and keeps the one with the **lower** $\ln\phi$ (lower Gibbs energy), which is the stable phase at that T and P. It reports "Vapor (3 roots, stable)" or "Liquid (3 roots, stable)", "Gas (T > Tc)" or "Supercritical fluid", and lists all real roots so the user can see the two-phase structure.

Results and Validation

The equations in the software were checked against the PR benchmark in Smith, Van Ness and Abbott (SVNA) for **n-butane at 500 K and 50 bar** (T_c = 425.1 K, P_c = 37.96 bar, ω = 0.200):

Quantity	SVNA (book)	This work	Difference
Z	0.6907	0.6909	0.03%
H^R (J/mol)	−4988	−4985	0.06%
S^R (J/(mol·K))	—	−7.4214	—
$\ln\phi$ / f (bar)	—	−0.30653 / 36.80	—

These are the values the software itself displays (Figs. 1–2), where it also reports V = 574.46 cm^3/mol and identifies the state as a supercritical fluid (T_r = 1.176, P_r = 1.317). The fourth-digit differences come from rounding of R and the critical constants in the book. **Two-phase check.** For n-butane at 350 K and 9.4573 bar the cubic has three physical roots: Z = 0.8081 (vapour, V = 2486 cm^3/mol), 0.1318 (unstable middle root) and 0.0366 (liquid, V = 112.6 cm^3/mol). The fugacity coefficients are almost equal (ϕ_v = 0.8374, ϕ_l = 0.8383), which shows that this pressure is just below the PR vapour pressure; the vapour root has the lower $\ln\phi$ and is selected as stable.

Fugacity consistency. Setting $f^v = f^l$ with the $\ln\phi$ expression derived above gives PR vapour pressures of 9.99 bar for propane (300 K), 9.47 bar for n-butane (350 K), 41.56 bar for CO_2 (280 K) and 10.53 bar for NH_3 (300 K). These agree with reference data (9.98, 9.47, 41.6 and 10.6 bar) to within 0.7%, which confirms the ϕ derivation.

Discussion and Conclusions

The software reproduces the textbook PR values for Z and H^R to the fourth significant figure, and the closed-form H^R, S^R and $\ln\phi$ give the right low-pressure limits and consistent phase equilibrium. For non-polar and weakly polar gases, PR is accurate enough for engineering work.

Limitations:

(a) **Near the critical point** ($T \approx T_c$, $P \approx P_c$), cubic equations over-predict Z and liquid volumes are less accurate (PR's Z_c = 0.307 against about 0.27 for real fluids).

(b) **Strongly polar or hydrogen-bonding fluids** (water, alcohols) are not well represented, because the standard $\alpha(T_r)$ depends only on T_r and ω.

(c) **Heat-capacity range:** the C_P^{ig} polynomials are fitted from 298.15 K upwards, so absolute H and S well below that temperature are extrapolations.

(d) **Pure components only:** mixtures would need mixing rules for a and b.

References: J. M. Smith, H. C. Van Ness, M. M. Abbott and M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill, 2018 (Secs. 3.6, 6.4, 13.6–13.7; Tables B.1, C.1). D.-Y. Peng and D. B. Robinson, *Ind. Eng. Chem. Fundam.* 15 (1976) 59–64.

Software Interface

The screenshots below show the software running the n-butane benchmark (500 K, 50 bar). The user selects a substance and an input mode (*T* & *P*, *T* & *V* or *P* & *V*), enters two state variables and taps *Calculate*; the PVT results, thermodynamic properties and graphs update together, and the results can be saved as a PDF.

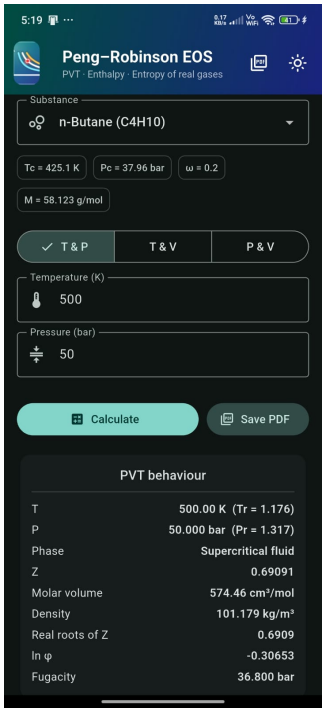


Fig. 1. Input screen and PVT results (n-butane, 500 K, 50 bar)

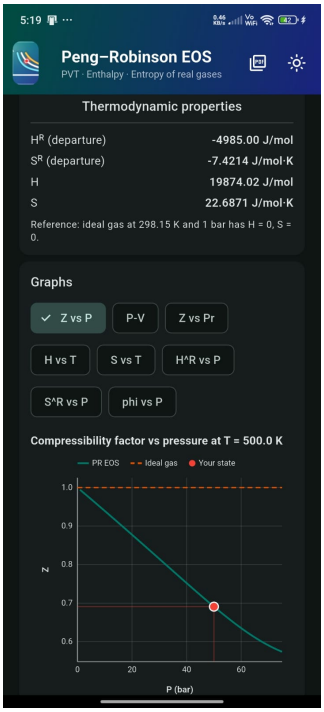


Fig. 2. Thermodynamic properties and Z vs P graph

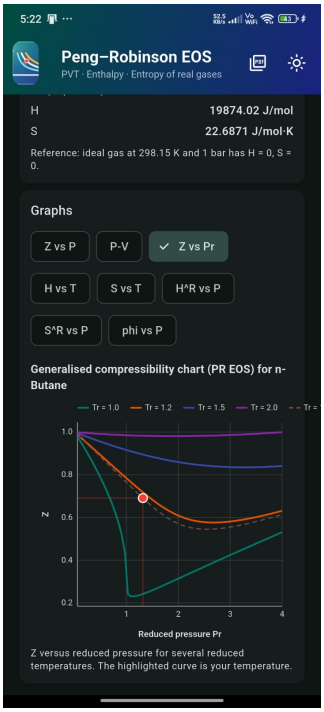


Fig. 3. Generalised compressibility chart (Z vs P_r)

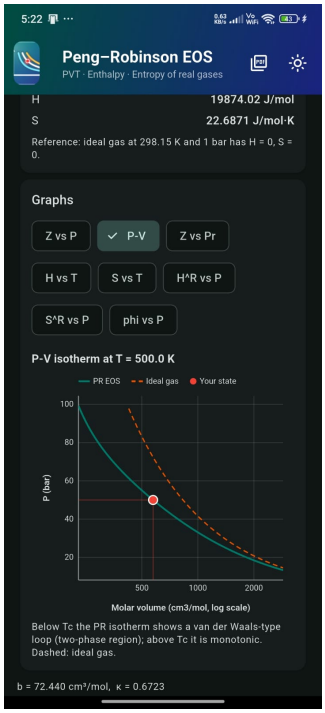


Fig. 4. P–V isotherm with ideal-gas comparison

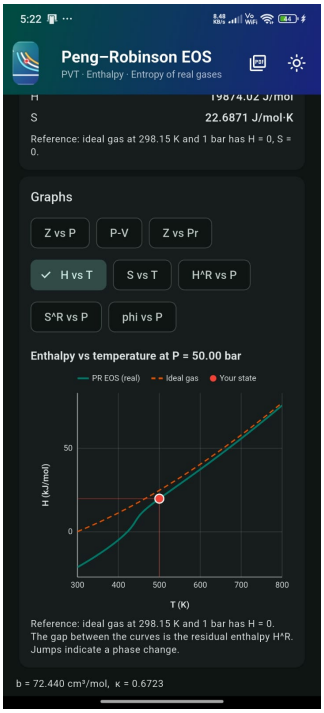


Fig. 5. Enthalpy vs temperature at 50 bar

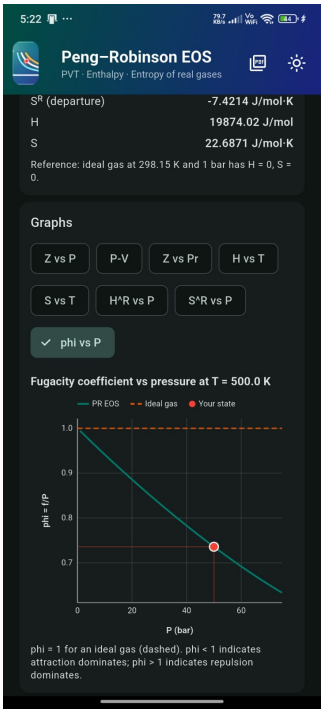


Fig. 6. Fugacity coefficient vs pressure at 500 K